Math-M-Addicts
MINDS THAT COUNT

SP250 Quiz 1

1. Definitions (15 points):
   a) (5 points): What is the definition of divisibility? (your answer should start with: Given integers a and b with $b \neq 0$....)
   b) (5 points): What is the definition of a prime number?
   c) (5 points): What is the definition of congruence in modular arithmetic? (Given integers a, b, m with $m \neq 0$, $a \equiv b \pmod{m}$ .....).

Solution:

a) Given integers a and b, we say a is divisible by b if and only if there exists another integer k such that $a = bk$.
b) A prime number is a natural number with exactly two distinct divisors, 1 and itself. Note that this definition excludes 1 as a prime number because 1 only has one distinct divisor.
c) Given integers a, b, m with $m \neq 0$, $a \equiv b \pmod{m}$ if and only if $m \mid (a - b)$.

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2. Using what you learned (50 points):

   a) (10 points): Prove that if $a \equiv b \pmod{m}$, then $10a \equiv 10b \pmod{m}$.
   b) (10 points): What is the 100th natural number not divisible by 3?
   c) (10 points): What is the sum of the first 100 natural numbers that are not divisible by 3?
   d) (10 points): What are the last three digits of $124997 \cdot 124996 \cdot 124994$?
   e) (10 points): Tanya and Yirong are playing a game of Nim with one pile of 45 stones, with Tanya going first. With each move, either one of them can take between 1 and 4 stones inclusively. The person who takes the last stone **wins**. Who has a winning strategy and why?

Solution:

a) $a \equiv b \pmod{m}$ means $m \mid (a - b)$, which implies $m \mid 10(a - b)$. Since $10(a - b) = 10a - 10b$, we have $m \mid (10a - 10b)$, which means $10a \equiv 10b \pmod{m}$ by definition of modular congruence.
b) Among every three consecutive natural numbers, two are not divisible by 3. So in the first 150 natural numbers, there are exactly 100 that are not divisible by 3. Thus, the 100th natural number not divisible by 3 is 149.
c) $1 + 2 + 4 + 5 + \cdots + 145 + 146 + 148 + 149 = (150 \cdot 100)/2 = 7500$
d) Taking everything modulo 1000, we get $124997 \cdot 124996 \cdot 124994 \equiv (-3) \cdot (-4) \cdot (-6) \equiv -72 \equiv 928 \pmod{1000}$. Thus, the last three digits are 928.
e) Yirong (the second player) has a winning strategy by always taking $5 - k$ stones where k is the number of stones taken by Tanya in the previous turn. This way, Yirong guarantees that the number of stones in the pile will be a multiple of 5 only after his turn. Since 0 is a multiple of 5, this means that Yirong is guaranteed to take the last stone.

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3. Know your proofs (15 points):

Prove that there are infinitely many prime numbers.

Solution:

We argue by way of contradiction. Assume there are finitely many primes and $\{p_1, p_2, p_3, ..., p_k\}$ is the set of all primes.

Now consider the number $N = p_1 \cdot p_2 \cdot p_3 \cdots p_k + 1$. N is not divisible by any of $p_1, p_2, p_3, ..., p_k$. However, all integers greater than 1 have at least one prime divisor. Thus, N must have a prime divisor that is not in $\{p_1, p_2, p_3, ..., p_k\}$, which contradicts the assumption that $\{p_1, p_2, p_3, ..., p_k\}$ is the set of all primes.

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4. Problem solving I (10 points):

Prove that there do not exist integers x and y such that

$$x^2 = 6y - 1.$$

Solution:

Let's consider both sides of the equation modulo 3.

For the left hand side, x^2 is either $0 \; or \; 1 \; (mod \; 3)$ as $0^2 \equiv 0, \; 1^2 \equiv 1, \; and \; 2^2 = 4 \equiv 1 \; (mod \; 3)$.

For the right hand side, $6y - 1 \equiv -1 \equiv 2 \; (mod \; 3)$.

Thus, it is impossible for the two sides of the equation to be equal as they are different modulo 3.

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5. Problem solving II (10 points):

A 2023x2023 square is given. Sergey and Tanya put 3x3 squares inside the big square in such a way that every 3x3 square fully covers exactly 9 squares of the 2023x2023 square. None of the 3x3 squares may overlap. The person who cannot put down a 3x3 square on their turn loses. Sergey goes first. Who wins with optimal play?

Solution:

Sergey (the first player) wins with optimal play. On his first move, Sergey places a 3x3 square in the center of the big square. After that, no matter what Tanya does on her turn, Sergey just places a 3x3 square at the location that is symmetric with respect to the center of the big square to that of the 3x3 square that Tanya had just placed, ensuring the big square is always symmetric with respect to its center after Sergey's move. This means that as long as Tanya is able to put down a 3x3 square, the symmetric location will be available for Sergey to put down a 3x3 square. Thus, Tanya will run out of moves first.